Chipmakers, Cloud Providers, and Frontier AI Labs:

Circular Investment and the Economics of the Compute Race

Nianzu Xiong

Department of Economics, University of Oxford

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Abstract

In this paper, we model the race for the AI frontier as an all-pay contest in which two labs buy compute from a supplier that earns a margin on every dollar they spend. The prize grows with both labs' spending through knowledge spillovers. Ownership links the layers: the supplier holds shares in the labs, and each lab's major investor holds shares in the supplier and possibly in the rival lab. We derive the symmetric equilibrium in closed form and show that circular holdings raise compute if and only if the lab's stake in the supplier, weighted by the margin, exceeds its stake in the rival weighted by one minus the spillover. The supplier's stake in a lab is neutral on its own, but it amplifies the lab's stake in the supplier. Unlike in R&D contests across industries, a lab's major investor can gain from a compute boom, because the supplier's margin is not competed away. The impossibility theorem reappears at the level of the ownership network: if the spillover plus the in-network margin is below one, no circular arrangement can raise compute and benefit its participants. We then show that exclusive deals discourage the rival lab and create value exactly when inclusive deals destroy it, that the labs' outside investors bear the losses from value-destroying booms, and that circularity makes compute more sensitive to margin compression. Finally, we add a racing layer of data-center and power developers.

JEL: D43, D44, G32, L13, L22, L63, O31. **Keywords:** circular investment, cross-shareholding, vendor financing, all-pay contests, spillovers, AI, compute.

1 Introduction

The frontier of artificial intelligence is being built on an unusual web of financial ties. Chip designers and cloud providers invest in the AI labs that buy their products. The labs commit to multi-year purchases of compute, and in some deals they receive equity or warrants in their suppliers that vest as purchases are made. For instance, Microsoft has been OpenAI's largest outside investor since 2019 and holds a stake of roughly 27% after OpenAI's 2025 recapitalization, while OpenAI buys large amounts of Azure compute. In October 2025, OpenAI received a warrant for AMD shares that vests as it deploys AMD GPUs. Investors and journalists call these arrangements "circular": money flows from the supplier to the lab as equity, back from the lab to the supplier as revenue, and sometimes back again as equity in the supplier (The Register, 2025; Morningstar, 2025).

Such arrangements raise two questions. Do they raise investment in compute? And do they create value, or do they inflate a compute boom whose costs fall on someone else? In this paper, we build a model to answer both. We model the race for the frontier as an all-pay contest: each lab spends on compute, and the lab that spends more reaches the frontier first and takes the prize, while both labs pay for their compute whether or not they win. The prize grows with total compute spending, which captures knowledge diffusion and market expansion. A supplier earns a margin on every dollar the labs spend. We use the terms "compute spending", "investment" and "effort" interchangeably throughout the paper.

The model builds on Xiong (2026), which models R&D races as all-pay contests between two industries, each consisting of two firms, with intra- and inter-industry spillovers in cost-reducing R&D and a Bertrand microfoundation for the prize. That paper studies symmetric cross-shareholdings among the four firms and proves an impossibility theorem: no symmetric cross-shareholding structure can raise a firm's R&D investment and its major investor's expected profit at the same time. Circular investment in the AI stack differs from that setting in two ways. First, the ties are *vertical*: the supplier does not race, and it earns a margin on every dollar the labs spend. Second, the ties are *asymmetric and partial*: a supplier may back one lab or all of them, and the labs are largely financed by outside investors. Both features turn out to be decisive.

We first derive the symmetric equilibrium investing strategy of each lab. The parameter design follows Xiong (2026): the major investor of each lab holds a share p of the rival lab (horizontal cross-shareholding) and a share q of the supplier (vertical cross-shareholding), and the supplier holds a share σ of each lab. We show that circular holdings raise each

lab's expected compute spending if and only if

$$q\mu > p(1-\alpha),$$

where μ is the supplier's margin and α is the spillover. This is the same condition as in Xiong (2026), with the margin μ in place of the inter-industry spillover. The supplier's stake σ does not appear in the condition. A one-way stake in a lab gives the lab cash, but it does not change the trade-off of the lab's major investor between the cost of compute and the chance of winning. What changes behavior is a link that pays the lab per dollar of compute, such as a stake in the supplier or a warrant that vests with purchases. Once such a link exists, the supplier's stake amplifies it.

The impossibility theorem of Xiong (2026) does not hold for a lab's major investor in this setting. When circular holdings raise compute, the lab's major investor also gains, because part of the extra spending comes back to him through the supplier's margin, and this margin is not competed away. The impossibility theorem reappears one level up. Margins are transfers within the ownership network, so the combined value of all firms linked by ownership is $V - (1 - \alpha - \mu_N)T$, where T is total compute and μ_N is the share of each compute dollar that is retained as profit inside the network. If $\alpha + \mu_N < 1$, no circular arrangement that raises compute can make its participants better off, even with side payments. If $\alpha + \mu_N > 1$, the labs under-invest from the network's point of view, because they pay a margin that is only a transfer, and every compute-raising arrangement creates value.

We then study four further questions. With outside investors, the major investors can profit from compute booms that destroy total value, and the losses fall on the labs' outside shareholders. An exclusive deal with one lab creates an asymmetric contest, which we solve in closed form; remarkably, the favored lab's expected spending does not change, and the deal works only by discouraging the rival. Exclusive deals therefore create value exactly when inclusive deals destroy it. Circularity raises the elasticity of compute spending with respect to the supplier's margin, so structures built in a high-margin environment can destroy value when margins are compressed. Finally, when a second layer of data-center and power developers also races, and the two layers' investments spill over onto each other, the value thresholds acquire cross-layer feedback terms, and bringing power producers into the ownership network can reverse the regime.

The rest of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the arrangements that motivate the model. Section 4 presents the model, and Section 5 gives a lemma on all-pay contests with linear payoffs. Sections 6–13 contain

the results for a single stack, and Section 14 adds data centers and power. Section 15 discusses welfare and policy, Section 16 gives an illustrative calibration, Section 17 lists testable implications, and Section 18 concludes.

2 Literature Review

Our paper uses the all-pay contest to model a race between firms. Hillman and Riley (1989) and Baye et al. (1996) characterize the equilibria of the all-pay auction with complete information, and Siegel (2009) gives a general characterization of payoffs in all-pay contests. Their papers serve as a foundation for subsequent work. Baye et al. (2012) study two-player contests in which each player's payoff depends linearly on both players' efforts, and our payoff structure is similar to theirs. In our model, the prize grows with total compute because of knowledge spillovers, and ownership adds further linear terms to each player's payoff. We give a lemma that covers all such linear payoffs and use it throughout the paper. The asymmetric equilibrium with an endogenous common prize in Section 9 is, to our knowledge, new in closed form.

Another main focus of our paper is cross-shareholding. Reynolds and Snapp (1986) study an industry in which firms hold shares in each other while competing in quantities, and show that partial cross-ownership raises prices. Gilo et al. (2006) analyze how partial cross-ownership between rivals facilitates tacit collusion in a repeated Bertrand game. Fu and Lu (2013) study cross-shareholdings in all-pay auctions without spillovers, and show that cross-shareholdings reduce effort. Xiong (2026) adds intra- and inter-industry spillovers to this setting and proves that no symmetric cross-shareholding structure can raise both R&D and the major investors' profits. Our paper differs from these papers in that the cross-holdings are vertical: one side of the network races, and the other side earns a margin on the racing side's spending. This is what allows a compute boom to benefit a lab's major investor, and it moves the impossibility theorem from the individual investor to the ownership network.

Our analysis of vertical holdings relates to the literature on partial vertical ownership and foreclosure. Flath (1989) studies vertical integration through shareholding interlocks, Greenlee and Raskovich (2006) study partial vertical ownership, and Levy et al. (2018) show how partial vertical integration can lead to foreclosure. Ordover et al. (1990) model foreclosure as raising rivals' costs. In our model, exclusive deals and vertical integration foreclose in a different way: they make the integrated lab a stronger competitor, which discourages the rival and lowers total input demand. Brennan et al. (1988) study vendor

financing, in which a supplier finances its customers' purchases; circular stakes can be viewed as a form of vendor financing in which the supplier's return comes partly through its margin.

Finally, our value thresholds relate to the comparison between noncooperative and cooperative R&D. d'Aspremont and Jacquemin (1988) and Kamien et al. (1992) show that firms that coordinate R&D invest more when spillovers are large. López and Vives (2019) show that overlapping ownership can raise R&D when technological spillovers are large, and Antón et al. (2021) provide evidence that common ownership can encourage innovation. Our thresholds generalize this comparison to vertical margins and to spillovers across layers of the AI stack.

3 Circular Investment in the AI Stack

This section describes the arrangements that motivate the model. Terms of individual deals change frequently and are often only partly disclosed, so the model is built to capture the *types* of linkage rather than any particular contract.

Supplier equity in labs. Chip designers and cloud providers hold equity in AI labs. Microsoft has been OpenAI's largest outside investor since 2019 and, after OpenAI's 2025 recapitalization, holds a stake of roughly 27% (Business Standard, 2025). In November 2025, Microsoft and Nvidia announced investments in Anthropic of up to $15 billion combined, alongside an Anthropic commitment to purchase $30 billion of Azure compute (Pure AI, 2025). Nvidia's equity investments in AI companies in 2026 have reportedly exceeded $40 billion, anchored by a $30 billion investment in OpenAI (The Next Web, 2026). In the model, these are the supplier's stakes σ.

Lab equity in suppliers. In October 2025, AMD and OpenAI announced a partnership to deploy six gigawatts of AMD GPUs, and OpenAI received a warrant for AMD shares that vests as deployment milestones are met (AMD, 2025). A warrant that vests with purchases is, economically, a volume rebate paid in the supplier's equity: the more the lab buys, the more of the supplier's margin it captures. In the model, these are the labs' stakes q.

Data centers and power. Data-center operators sit between chipmakers and labs. Nvidia has invested repeatedly in CoreWeave, including a further $2 billion in January 2026

(Bloomberg, 2026). The result is a chain of stakes and purchase commitments running from chips through data centers to labs and back (The Register, 2025). The binding constraint on new data centers is increasingly electricity. Hyperscalers have signed long-term power purchase agreements and made commitments to nuclear and other firm generation. Most of these are contracts rather than equity, and their realism has been questioned (Carnegie, 2026). In the model, power is a layer whose margins are either inside or outside the ownership network.

Commentators disagree on whether circularity is benign (risk sharing, financing of a capital-intensive buildout) or a source of fragility (revenue round-tripping, inflated demand) (Morningstar, 2025; The Register, 2025). Our model gives conditions under which each view is right. Figure 1 summarizes the links described above.

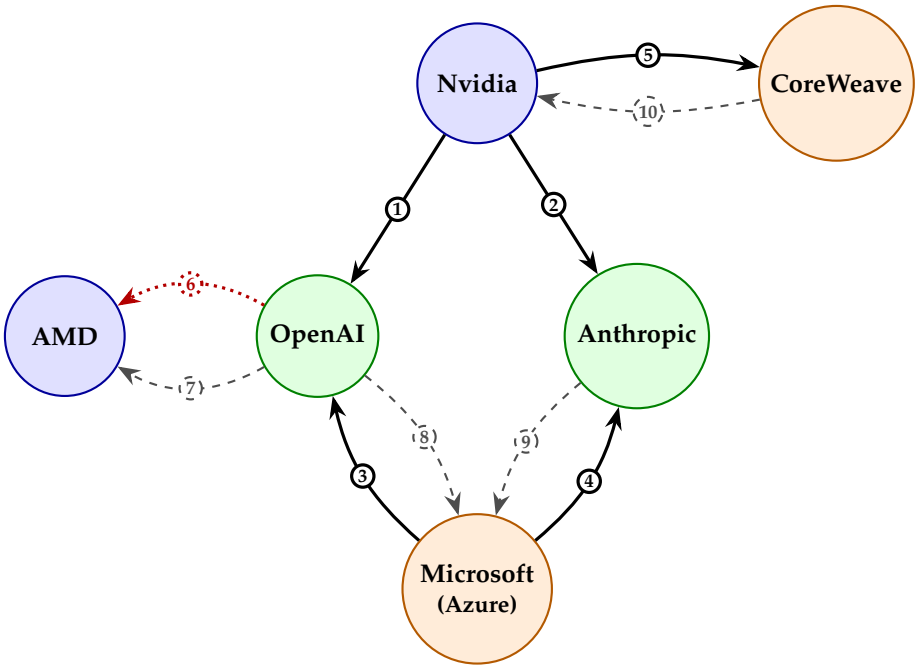


Figure 1: Selected circular links in the AI stack, based on the public reports cited in Section 3. Blue circles are chipmakers, orange circles are cloud and data-center providers, and green circles are frontier labs. The map is illustrative and not exhaustive. In the model, solid arrows from suppliers into labs correspond to σ, dotted arrows from labs into suppliers to q, and dashed arrows to the spending e_i on which suppliers earn the margin μ.

4 The Model

4.1 The Baseline Model

There are two AI labs, a and b, a compute supplier S, which represents the chips-and-cloud layer, and an outside layer O, which represents, for example, independent power producers. Section 14 adds a second layer that races.

The timing of the model is as follows. Ownership is in place at the start. The two labs then simultaneously choose compute spending $e_i \in [0, \infty)$, measured in dollars. The lab that spends more reaches the frontier first and earns the prize

$$R(e_a, e_b) = V + \alpha\,(e_a + e_b), \tag{1}$$

while the other lab earns nothing from the race. In the event of a tie, each lab wins with probability $\frac{1}{2}$. Here $V > 0$ is the exogenous value of leadership, that is, the rents of the frontier model over its product cycle, and $\alpha(e_a + e_b)$ captures the increase in the prize from both labs' compute. Some of the loser's work diffuses to the winner through publications, staff mobility, distillation or open weights, and both labs' spending expands the market for frontier AI. We assume $0 < \alpha < 1$; without this assumption, the marginal reward from spending could exceed its cost, and the labs would want to spend without bound.

Each dollar of lab spending generates profit $\mu \in [0, 1)$ for the supplier S and profit $\mu_O \geq 0$ for the outside layer O. The remaining $1 - \mu - \mu_O > 0$ is the real resource cost of compute. The firms' profits are therefore

$$\Pi_i = R \cdot \mathbf{1}\{i \text{ wins}\} - e_i, \qquad \Pi_S = \mu(e_a + e_b), \qquad \Pi_O = \mu_O(e_a + e_b). \tag{2}$$

We take the margins as given in the baseline model and study changes in them in Section 10.

4.2 Cross-Shareholding

Each lab is controlled by a *major investor*, who appoints a manager to make compute decisions on his behalf. The supplier is controlled by its own major investor. Following Xiong (2026), we focus on symmetric holdings. The major investor of lab i owns a share θ of lab i, a share p of the rival lab j, and a share q of the supplier. The supplier's major investor owns a share σ of each lab. Passive outside investors own a share ϕ_L of each lab and a share ϕ_S of the supplier. The ownership table is

	Lab i	Lab j	Supplier S
Major investor of lab i	θ	p	q
Major investor of the supplier	σ	σ	$1-2q-\phi_S$
Outside investors	ϕ_L	ϕ_L	ϕ_S

Each column sums to one, so $\theta = 1 - p - \sigma - \phi_L$. We allow the major investors to control their firms even when their economic stake is below one half, as is common for founder-led labs and for firms with dual-class shares. As in Xiong (2026), we call p the *horizontal* cross-shareholding and q the *vertical* cross-shareholding; we call (σ, q) the *circular* holdings and (ϕ_L, ϕ_S) *outside finance*. The benchmark has no holdings: $\sigma = q = p = \phi_L = \phi_S = 0$ and $\theta = 1$.

The share q stands for any arrangement through which a lab's major investor captures part of the supplier's margin: equity, warrants that vest with purchases, or revenue-share rebates. Section 7 shows that purchase commitments and other instruments are evaluated by the same value test.

The *ownership network* is the set of major investors. Its *in-network margin* μ_N is the share of each dollar of lab spending that becomes profit for a firm inside the network. In the baseline model, $\mu_N = \mu$, since the supplier is inside the network and the outside layer is not. Figure 2 depicts the structure.

Consider the problem of the major investor of lab i. He receives a share θ of lab i's profit, a share p of lab j's profit and a share q of the supplier's profit:

$$W_i = \theta\,\Pi_i + p\,\Pi_j + q\,\Pi_S. \tag{3}$$

If lab i wins, he gets a θ share of the payoff $V + \alpha(e_i + e_j) - e_i$, a p share of the rival's losing cost $-e_j$, and a q share of the supplier's profit $\mu(e_i + e_j)$. If lab i loses, he gets a θ share of the losing cost $-e_i$, a p share of the rival's winning payoff $V + \alpha(e_i + e_j) - e_j$, and again a q share of the supplier's profit. Notice that his payoff from the supplier is the same whichever lab wins. So the major investor's payoff is

$$W_i(e_i, e_j) = \begin{cases} \theta[V + \alpha(e_i + e_j) - e_i] - pe_j + q\mu(e_i + e_j) & \text{if } e_i > e_j, \\ -\theta e_i + p[V + \alpha(e_i + e_j) - e_j] + q\mu(e_i + e_j) & \text{if } e_i < e_j, \end{cases}$$

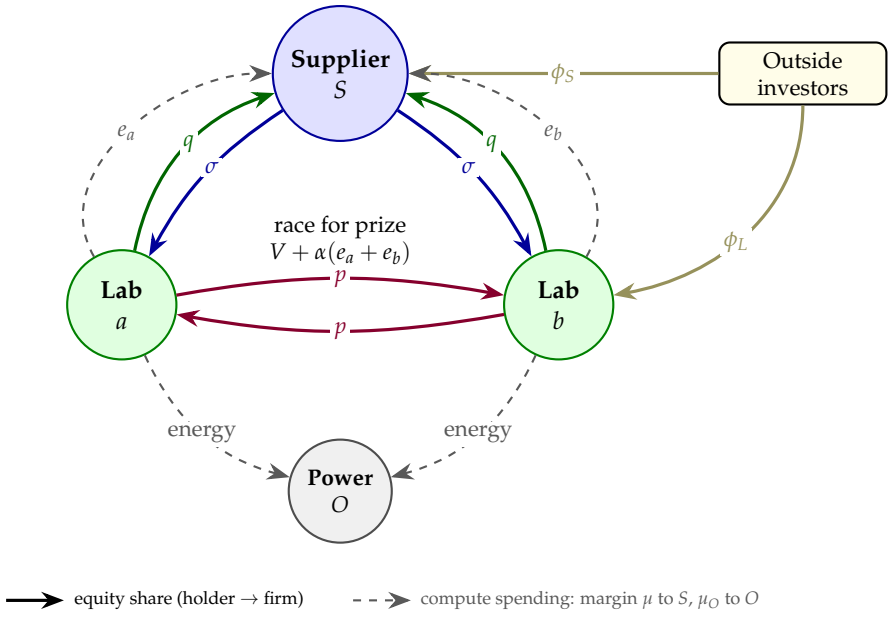


Figure 2: Ownership and cash flows in the model. The supplier's major investor holds σ of each lab; the major investor of each lab holds q of the supplier and p of the rival lab; outside investors hold ϕ_L of each lab and ϕ_S of the supplier (only one ϕ_L arrow is drawn). Each dollar of lab spending yields the margin μ to the supplier and μ_O to the outside power layer.

with the average of the two in the event of a tie. Rearranging terms, we get

$$W_i(e_i, e_j) = \begin{cases} \theta V - [\theta(1-\alpha) - q\mu]\, e_i - [p - \theta\alpha - q\mu]\, e_j & \text{if } e_i > e_j, \\ pV - [\theta - p\alpha - q\mu]\, e_i - [p(1-\alpha) - q\mu]\, e_j & \text{if } e_i < e_j. \end{cases} \tag{4}$$

This has exactly the structure of the investor's payoff in Xiong (2026), with the margin μ in place of the inter-industry spillover β and the own share θ in place of $1 - p - 2q$. The supplier's major investor receives $W_S = (1 - 2q - \phi_S)\Pi_S + \sigma(\Pi_a + \Pi_b)$. Product-market conduct after the race is summarized by V and does not depend on ownership, which isolates the investment channel; Section 15 discusses conduct effects.

To focus on the interesting case where each lab makes positive expected compute investment, we need two assumptions:

$$\theta > p, \tag{A1}$$

$$\theta(1-\alpha) > q\mu. \tag{A2}$$

Assumption (A1) means that the major investor of a lab prefers his own lab to win the race,

so that the value of winning over losing, $(\theta - p)V$, is positive. Assumption (A2) means that the major investor's share of the cost of a dollar of compute that is not recovered through the prize, $\theta(1-\alpha)$, exceeds what he recovers through the supplier, $q\mu$. It will be clear from our later calculations that (A2) is critical for expected spending to be finite.

5 A Lemma on All-Pay Contests with Linear Payoffs

The payoff (4) is linear in both labs' spending, with different coefficients depending on whether the lab wins or loses. All of our results rest on the following lemma, which gives the symmetric equilibrium of any such contest.

Lemma 1 (Linear contests)**.** *Two players choose efforts $e_i, e_j \geq 0$. If $e_i > e_j$, player i's payoff is $w_0 + w_1 e_i + w_2 e_j$; if $e_i < e_j$, it is $\ell_0 + \ell_1 e_i + \ell_2 e_j$; ties are split. Suppose*

$$\Delta \equiv w_0 - \ell_0 > 0, \qquad d \equiv w_1 - \ell_1 = w_2 - \ell_2 > 0, \qquad \kappa \equiv -\ell_1 > d.$$

Then there is a symmetric equilibrium in which each player randomizes according to the c.d.f.

$$F(x) = \frac{\kappa}{d}\left[1 - \left(\frac{\Delta}{\Delta + 2dx}\right)^{1/2}\right] \tag{5}$$

over the interval

$$\left[0,\ \frac{\Delta}{2d}\left\{\left(\frac{\kappa}{\kappa - d}\right)^{2} - 1\right\}\right]. \tag{6}$$

Each player's expected effort and expected equilibrium payoff are

$$m = \frac{\Delta}{2(\kappa - d)}, \qquad U^{*} = \ell_0 + \ell_2\, m. \tag{7}$$

Proof. Let b denote the upper bound in (6), which is derived by solving $F(b) = 1$. Let f be the p.d.f. of F, and write $u(x) = \Delta + 2dx$. Then

$$f(x) = \kappa\, \Delta^{1/2} u(x)^{-3/2}, \qquad f(x)\, u(x) = \kappa \left(\frac{\Delta}{u(x)}\right)^{1/2}, \qquad d\,F(x) = \kappa - \kappa \left(\frac{\Delta}{u(x)}\right)^{1/2}.$$

Suppose player j follows F while player i deviates to investing $s \in [0, b]$. Then player i's

expected payoff is

$$U(s) = \int_0^s \big(w_0 + w_1 s + w_2 y\big) f(y)\, dy + \int_s^b \big(\ell_0 + \ell_1 s + \ell_2 y\big) f(y)\, dy.$$

Differentiating with respect to s and using $w_0 - \ell_0 = \Delta$ and $w_1 - \ell_1 = w_2 - \ell_2 = d$,

$$U'(s) = f(s)\big[\Delta + 2ds\big] + w_1 F(s) + \ell_1\big[1 - F(s)\big] = f(s)\, u(s) + d\, F(s) - \kappa = 0,$$

by the expressions above. So $U(s)$ is constant for all $s \in [0, b]$. Deviating to $s > b$ makes player i win with probability one and yields $w_0 + w_1 s + w_2 m$, which is strictly decreasing in s because $w_1 = \ell_1 + d = -(\kappa - d) < 0$. Thus randomizing according to F is optimal.

Now we compute expected effort:

$$m = \int_0^b \big[1 - F(x)\big] dx = \Big(1 - \frac{\kappa}{d}\Big) b + \frac{\kappa}{d} \int_0^b \left(\frac{\Delta}{\Delta + 2dx}\right)^{1/2} dx.$$

Since $u(b)^{1/2} = \Delta^{1/2}\kappa/(\kappa - d)$, the integral equals $(\Delta^{1/2}/d)\big[u(b)^{1/2} - \Delta^{1/2}\big] = \Delta/(\kappa - d)$. Using $b = \Delta(2\kappa - d)/[2(\kappa - d)^2]$,

$$m = -\frac{\Delta(2\kappa - d)}{2d(\kappa - d)} + \frac{\kappa\Delta}{d(\kappa - d)} = \frac{\Delta}{2(\kappa - d)}.$$

Finally, since F has no atom, a player who invests zero loses with probability one, so $U^* = U(0) = \ell_0 + \ell_2 m$. □

The lemma has a simple economic reading. $\Delta + d(e_i + e_j)$ is the value of winning over losing, and $\kappa - d$ is the net marginal cost of effort: the marginal cost when losing, less the part recovered through the prize. Expected effort is half the intercept of the value of winning divided by the net marginal cost. A player's expected payoff equals what it earns when it loses for sure, because the rents from winning are competed away.

6 Equilibrium: It Takes Two to Circle

Now we derive the equilibrium of the race with cross-shareholding. As in Xiong (2026), we denote the expected winning prize and the expected losing prize of a lab's major investor

by V^w and V^l, and their difference by $\tilde{V}$:

$$V^w = \theta V, \qquad V^l = pV, \qquad \tilde{V} = V^w - V^l = (\theta - p)V.$$

Theorem 1 (Equilibrium with circular holdings)**.** *Suppose* (A1) *and* (A2) *hold. Then the symmetric equilibrium investing strategy of each lab is to randomize according to the c.d.f.*

$$B(x) = \frac{\theta - p\alpha - q\mu}{(\theta - p)\alpha}\left[1 - \left(\frac{\tilde{V}}{\tilde{V} + 2(\theta - p)\alpha x}\right)^{1/2}\right]$$

over the interval

$$\left[0,\ \frac{V}{2\alpha}\left\{\left[\frac{\theta - p\alpha - q\mu}{\theta(1-\alpha) - q\mu}\right]^2 - 1\right\}\right].$$

Each lab's expected compute spending is

$$\bar{e} = \frac{(\theta - p)\,V}{2\left[\theta(1-\alpha) - q\mu\right]} = \frac{(1 - p/\theta)\,V}{2\,(1 - \alpha - r)}, \qquad r \equiv \frac{q\mu}{\theta}, \tag{8}$$

and the expected payoff of each lab's major investor is

$$\mathbb{E}W_i = pV + \left[q\mu - p(1-\alpha)\right]\bar{e}. \tag{9}$$

In the benchmark, each lab randomizes according to $G(x) = \frac{1}{\alpha}\left[1 - \left(\frac{V}{V+2\alpha x}\right)^{1/2}\right]$ *over* $\left[0, \frac{(2-\alpha)V}{2(1-\alpha)^2}\right]$*, expected spending is* $\bar{e}^0 = V/[2(1-\alpha)]$*, each lab expects zero profit, and the supplier expects* $\mu V/(1-\alpha)$*.*

Proof. By (4), the coefficients of Lemma 1 are $w_0 = \theta V$, $w_1 = -[\theta(1-\alpha) - q\mu]$, $w_2 = -[p - \theta\alpha - q\mu]$, $\ell_0 = pV$, $\ell_1 = -[\theta - p\alpha - q\mu]$ and $\ell_2 = -[p(1-\alpha) - q\mu]$. Hence $\Delta = (\theta - p)V = \tilde{V} > 0$ by (A1), $w_1 - \ell_1 = w_2 - \ell_2 = (\theta - p)\alpha = d > 0$, and $\kappa = \theta - p\alpha - q\mu$, so that $\kappa - d = \theta(1-\alpha) - q\mu > 0$ by (A2). Plugging these into Lemma 1 gives the c.d.f., the interval, $\bar{e}$ and $\mathbb{E}W_i = \ell_0 + \ell_2\bar{e}$. The benchmark is the case $\theta = 1$, $p = q = 0$. □

We call $r = q\mu/\theta$ the *effective rebate*. For each dollar a lab spends, its major investor bears θ dollars of the cost, while the rest falls on the lab's other shareholders, and he recovers $q\mu$ dollars through his stake in the supplier. Relative to his own share of the cost, he gets back r per dollar.

The following result is the analogue of Proposition 4 of Xiong (2026).

Proposition 1 (When circular holdings raise compute)**.** *Circular holdings raise each lab's expected compute spending above the benchmark if and only if*

$$q\mu > p(1-\alpha). \tag{10}$$

The condition does not depend on the supplier's stake σ or on outside finance ϕ_L.

Proof. By (8), $\bar{e} > \bar{e}^0$ if and only if $(\theta - p)(1-\alpha) > \theta(1-\alpha) - q\mu$, which is equivalent to $q\mu > p(1-\alpha)$. □

We pay special attention to the case where labs hold no shares in each other, $p = 0$.

Observation 1 (It takes two to circle)**.** *Let $p = 0$.*

(a) One-way stakes are neutral. *If $q = 0$, then $\bar{e} = \bar{e}^0$ for every σ and ϕ_L.*

(b) The vertical stake is a rebate. *$\bar{e}$ is strictly increasing in q, and strictly increasing in μ whenever $q > 0$.*

(c) Amplification. *If $q > 0$, then $\bar{e}$ is strictly increasing in σ and in ϕ_L, because both shrink θ and raise r.*

(d) Multiplier. *$\bar{e}/\bar{e}^0 = (1-\alpha)/(1-\alpha-r)$, which grows without bound as r approaches $1-\alpha$.*

When $p > 0$, $\bar{e}$ is strictly decreasing in p.

Proof. This result follows immediately from (8). □

The intuition behind part (a) is rather clear. A supplier that takes a passive stake in a lab gives the lab cash, but it does not change the major investor's trade-off between the cost of compute and the chance of winning, because both are scaled by the same share θ. What changes behavior is a link that pays the lab *per dollar of compute*: a stake in the supplier, a warrant that vests with purchases, or a rebate. Once such a link exists, the supplier's stake in the lab, as well as the lab's reliance on outside capital, magnifies it, because the major investor's own share of the cost falls while his share of the supplier's margin does not. Circularity is therefore multiplicative. The horizontal holding p works in the opposite direction: as in Fu and Lu (2013) and Xiong (2026), holding shares in one's rival makes competition less intense, because hurting the rival hurts oneself.

Now we analyze the effect of circular holdings on the payoffs of the lab's major investor. In Xiong (2026), the condition under which cross-shareholding raises R&D is precisely the opposite of the condition under which it raises the major investor's profit, which gives the impossibility theorem. Here the situation is different.

Proposition 2 (Accounting losses, investor gains)**.** *Each lab's expected accounting profit is*

$$\mathbb{E}\Pi_i = \frac{V}{2} - (1-\alpha)\,\bar{e},$$

which is negative whenever $\bar{e} > \bar{e}^0$*. The supplier's expected profit is* $2\mu\bar{e}$*. If* (10) *holds, then the major investor of each lab expects a strictly positive payoff,* $\mathbb{E}W_i > pV \geq 0$*. In particular, with* $p = 0$ *and* $q > 0$*,*

$$\mathbb{E}\Pi_i = -\frac{r\,V}{2(1-\alpha-r)} < 0, \qquad \mathbb{E}W_i = q\mu\,\bar{e} > 0.$$

Proof. Exactly one lab wins, so $\mathbb{E}\Pi_i = \frac{1}{2}\mathbb{E}R - \bar{e} = \frac{V}{2} + \alpha\bar{e} - \bar{e}$. The rest follows from (8) and (9). □

Observation 2. *The impossibility theorem of Xiong (2026) does not hold for the major investors of the labs: whenever circular holdings raise compute, they also raise each lab's major investor's expected payoff.*

The reason is that in Xiong (2026), the firms in both industries race, and each firm expects zero profit in equilibrium without cross-shareholding, so the rents in the other industry are competed away as well. A major investor's stake in the other industry therefore pays off only through the spillover β, which the race in that industry also dissipates. Here the supplier does not race, and its margin on the extra compute is not competed away. Part of the margin flows back to the lab's major investor through his stake q. Circular structures therefore produce a recognizable pattern: labs that lose money on paper, major investors who do well through their claims on the supplier, and a supplier whose revenue rises with the degree of circularity. Whether the system as a whole gains is a separate question, which we answer in the next section.

Example 1. Let $V = 1$, $\alpha = 0.2$ and $\mu = 0.6$. In the benchmark, each lab randomizes according to

$$G(x) = 5 - 5\Big[\frac{1}{1+0.4x}\Big]^{1/2}$$

over $[0, 1.406]$, spends $\bar{e}^0 = 0.625$ in expectation and earns zero, and the supplier earns 0.75. Now let the supplier take $\sigma = 0.3$ of each lab and let each lab's major investor receive

$q = 0.2$ of the supplier, with $p = \phi_L = 0$. Then $\theta = 0.7$, $r = 0.2 \cdot 0.6/0.7 = 0.171$, and by Theorem 1 the equilibrium investing strategy becomes

$$B(x) = 4.143 - 4.143\left[\frac{1}{1 + 0.4x}\right]^{1/2}$$

over $[0, 1.844]$. Figure 3 plots both c.d.f.s. Notice that G is first-order stochastically dominated by B: with circular holdings, labs tend to spend more. Expected spending rises to $\bar{e} = 0.795$, an increase of 27%. Each lab's expected accounting profit is -0.136, while each lab's major investor earns $q\mu\bar{e} = 0.095$. The supplier's profit rises to 0.955, of which its major investor retains 60%, in addition to its share σ of the labs' losses. The combined value of the network falls from 0.75 to 0.682, because $\alpha + \mu = 0.8 < 1$ (Theorem 2).

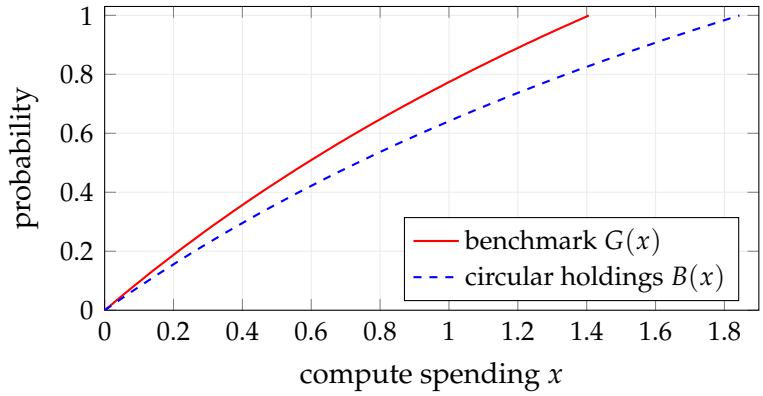


Figure 3: Equilibrium investing strategies in Example 1 ($V = 1$, $\alpha = 0.2$, $\mu = 0.6$; circular holdings $\sigma = 0.3$, $q = 0.2$).

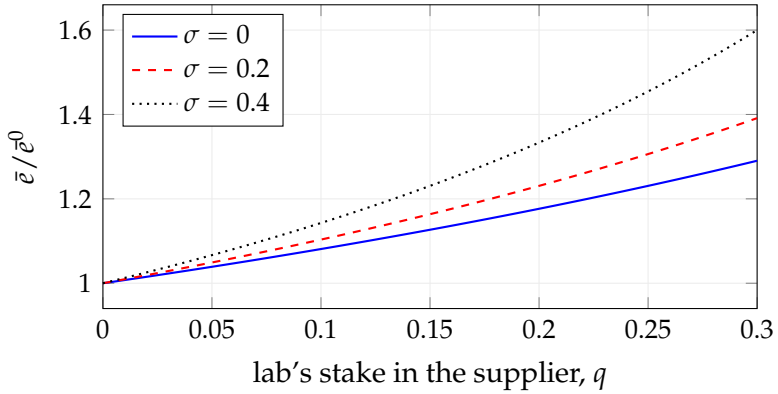


Figure 4: The circularity multiplier (Observation 1), with $\alpha = 0.2$, $\mu = 0.6$ and $p = \phi_L = 0$. With $q = 0$, the supplier's stake σ has no effect. With $q > 0$, σ amplifies the effect of q.

7 Value: The Impossibility Theorem for the Network

7.1 The Network-Value Identity

Consider the network of major investors, and suppose for now that there is no outside finance ($\phi_L = \phi_S = 0$). Ownership is then closed: every share of every firm in the network is held by a major investor. The network's combined payoff is the sum of the firms' profits, $J = \Pi_a + \Pi_b + \Pi_S$.

Theorem 2 (The circular-investment impossibility theorem)**.** *Let $T = \mathbb{E}(e_a + e_b)$ be expected total compute. For any strategy profile and any arrangement among members of the network that leaves V, α and μ unchanged, including equity stakes, warrants, rebates, purchase commitments and side payments,*

$$\mathbb{E}J = V - (1 - \alpha - \mu_N)\, T. \tag{11}$$

Hence, relative to the benchmark:

(a) Impossibility. *If $\alpha + \mu_N < 1$, every arrangement that raises total compute lowers the network's value. There is no arrangement that can simultaneously raise compute and make all participants better off, even with side payments.*

(b) Possibility. *If $\alpha + \mu_N > 1$, every arrangement that raises total compute raises the network's value, and with side payments it can make all participants better off.*

Proof. Exactly one lab wins, so $\Pi_a + \Pi_b = R - e_a - e_b$ for every strategy profile. Adding $\Pi_S = \mu_N(e_a + e_b)$ and taking expectations gives (11). Side payments only redistribute J, so all participants can gain if and only if J rises, which requires $(\alpha + \mu_N - 1)\Delta T > 0$. □

The identity is an accounting fact, but it organizes everything that follows. From the network's point of view, a dollar of compute costs $1 - \mu_N$, because the margin is a transfer to the supplier, and returns α through a larger prize. The labs, however, pay the full dollar. When $\alpha + \mu_N < 1$, the rent-dissipating race already over-invests from the network's point of view, and any circular arrangement that raises compute makes things worse. This is the analogue, for vertical networks, of the impossibility theorem in Xiong (2026): the impossibility is no longer about each major investor, but about the network as a whole. When $\alpha + \mu_N > 1$, the margin is a wedge: labs under-invest because they pay a markup that is only a transfer within the network. Circular arrangements then correct a double-marginalization problem in compute, and the network gains.

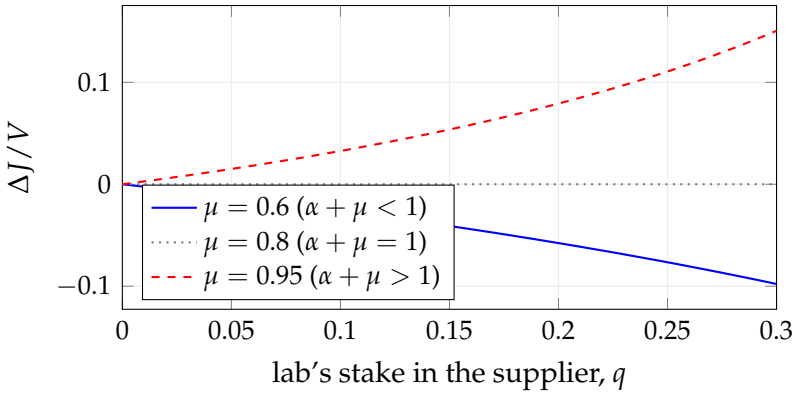


Figure 5: Change in network value from a circular structure with $\sigma = 0.2$ and $p = 0$, as a function of q ($\alpha = 0.2$). The sign is determined by $\alpha + \mu - 1$ (Theorem 2).

7.2 Instrument Irrelevance

Theorem 2 does not depend on how compute is raised. To illustrate, consider a take-or-pay purchase commitment, a common feature of supplier deals.

Proposition 3 (Purchase commitments)**.** *Without holdings, suppose each lab commits to pay at least $C > 0$ for compute, so that usage up to C costs nothing at the margin. Then each lab spends $\bar{e} = \bar{e}^0 + C/(1-\alpha)$ in expectation, its expected accounting profit is $-C$, and network value changes by $-2(1-\alpha-\mu_N)\,C/(1-\alpha)$.*

Proof. Write $e_i = C + x_i$ with $x_i \geq 0$. The race over x has prize $V' + \alpha(x_a + x_b)$ with $V' = V + 2\alpha C$ and unit marginal cost. By Lemma 1, $\mathbb{E}x = V'/[2(1-\alpha)]$, so $\bar{e} = C + V'/[2(1-\alpha)] = \bar{e}^0 + C/(1-\alpha)$. The race over x dissipates its rents fully, which leaves a profit of $-C$. Applying (11) gives the change in network value. □

A commitment of C raises spending by more than C because of spillovers: each lab's committed compute raises the prize for which both labs compete. Commitments, equity rebates and warrants are therefore different instruments with the same value test. A commitment that comes bundled with an equity investment ("we invest, you commit to our cloud") creates network value if and only if $\alpha + \mu_N > 1$; the equity part only splits the surplus or the loss.

8 Who Pays? Outside Investors

AI labs are financed largely by outside investors, and suppliers are publicly held. With outside finance, the major investors' combined payoff is $J_{\text{ins}} = J - \phi_L(\Pi_a + \Pi_b) - \phi_S \Pi_S$.

Theorem 3 (Major investors versus outside investors)**.** *Let $p = 0$. A circular structure that raises spending per lab by $\Delta\bar{e} > 0$ changes the major investors' combined payoff by*

$$\Delta J_{\text{ins}} = 2\Delta\bar{e}\left[\mu(1-\phi_S) - (1-\alpha)(1-\phi_L)\right]. \tag{12}$$

The labs' outside investors lose $2\phi_L(1-\alpha)\Delta\bar{e}$, and the supplier's outside investors gain $2\phi_S\mu\Delta\bar{e}$. In particular, if

$$\frac{(1-\alpha)(1-\phi_L)}{1-\phi_S} < \mu < 1-\alpha,$$

the structure is profitable for the major investors and destroys total value, and the net loss is borne by the labs' outside investors.

Proof. By Theorem 2 and Proposition 2, $\Delta J = 2(\alpha+\mu-1)\Delta\bar{e}$, $\Delta\mathbb{E}\Pi_i = -(1-\alpha)\Delta\bar{e}$ and $\Delta\mathbb{E}\Pi_S = 2\mu\Delta\bar{e}$. Then $\Delta J_{\text{ins}} = \Delta J + 2\phi_L(1-\alpha)\Delta\bar{e} - 2\phi_S\mu\Delta\bar{e}$, which rearranges to (12). □

Theorem 3 explains why circular structures can be attractive to those who design them even when they destroy value. The labs' accounting losses are shared with outside investors, while the margin they generate flows to the supplier and, through q, back to the labs' major investors. Outside finance enters twice: it raises the rebate $r = q\mu/\theta$ (Observation 1(c)), and it shifts losses onto outsiders. The model thus identifies a channel through which a compute boom can be privately rational and collectively wasteful. It also identifies who bears the risk: the outside investors in the labs, not the suppliers' shareholders, who benefit from higher volume.

9 Pick a Horse or Back the Field

Some suppliers invest in several competing labs, while others tie themselves to one. In the model, an *inclusive* deal gives both labs the same holdings, and an *exclusive* deal links the supplier to lab a only. With $p = 0$, the major investor of lab a then has the rebate $r = q\mu/\theta$, so his normalized marginal cost of compute is $c \equiv 1 - r \in (\alpha, 1)$, while lab b has marginal cost 1. The exclusive deal turns the race into an asymmetric contest with a common endogenous prize.

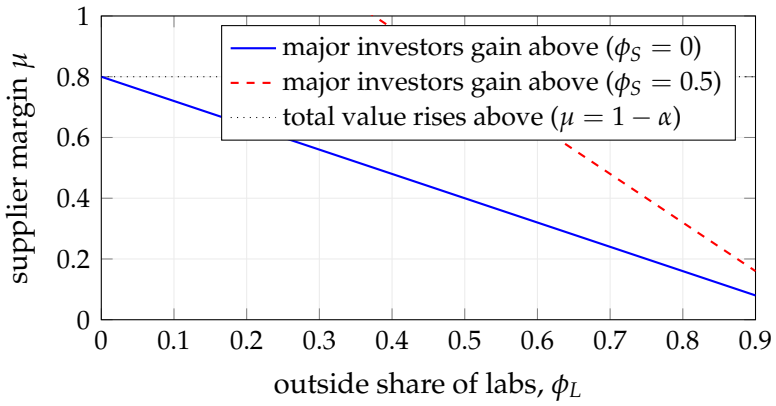


Figure 6: Theorem 3 with $\alpha = 0.2$. Between the solid line and the dotted line ($\phi_S = 0$), compute-raising circular structures benefit the major investors but destroy total value, at the expense of the labs' outside investors.

Lemma 2 (Asymmetric linear contest)**.** *Two players compete for the prize $V + \alpha(e_1 + e_2)$, with $\alpha > 0$, at marginal costs $c_1 < c_2$, where $c_1 > \alpha$. Write $u(x) = V + 2\alpha x$. There is an equilibrium in which both players randomize over $[0, \bar{x}]$, with*

$$\bar{x} = \frac{V}{2\alpha}\left[\left(\frac{c_2}{c_2 - \alpha}\right)^2 - 1\right],$$

according to

$$G_1(x) = \frac{c_2}{\alpha}\left[1 - \left(\frac{V}{u(x)}\right)^{1/2}\right], \qquad G_2(x) = \frac{c_1}{\alpha} - \frac{(c_1 - \alpha)c_2}{\alpha(c_2 - \alpha)}\left(\frac{V}{u(x)}\right)^{1/2}.$$

The weak player has an atom at zero of mass $G_2(0) = \frac{c_2 - c_1}{c_2 - \alpha}$. Moreover:

(i) expected efforts are $\mathbb{E}e_1 = \dfrac{V}{2(c_2 - \alpha)}$ and $\mathbb{E}e_2 = \dfrac{(c_1 - \alpha)\,V}{2(c_2 - \alpha)^2}$;

(ii) the strong player wins with probability $\dfrac{2c_2 - c_1 - \alpha}{2(c_2 - \alpha)}$;

(iii) the expected payoffs are $V\dfrac{c_2 - c_1}{c_2 - \alpha}$ for the strong player and 0 for the weak player.

Proof. Please see Appendix A. □

Two features of Lemma 2 drive the results. First, the strong player's expected effort depends only on the *weak* player's cost, because in equilibrium G_1 must keep the weak

player indifferent. Second, the weak player responds to its disadvantage by staying out with positive probability.

Figure 7 illustrates. With $c_2 = 1$, the strong player's distribution G_1 coincides *exactly* with the symmetric benchmark: a lab that receives an exclusive rebate behaves as if nothing had changed. The rival's distribution shifts down. It places mass $(1 - c_1)/(1 - \alpha)$ at zero and randomizes over the same interval with lower density. All of the deal's effect on behavior therefore operates through the rival.

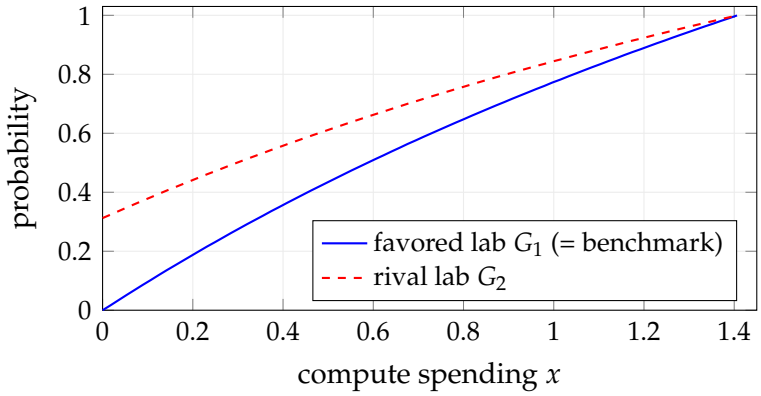


Figure 7: Equilibrium investing strategies under an exclusive deal (Lemma 2), with $V = 1$, $\alpha = 0.2$, $c_1 = 0.75$ and $c_2 = 1$. The favored lab's distribution equals the symmetric benchmark. The rival stays out with probability 0.31, and the favored lab wins with probability 0.66.

Theorem 4 (Exclusive versus inclusive deals)**.** *Fix an effective rebate $r \in (0, 1 - \alpha)$ and let $c = 1 - r$.*

(a) Exclusive deal. *The favored lab's expected spending is unchanged at $\bar{e}^0 = V/[2(1 - \alpha)]$. The rival's falls to $(c - \alpha)V/[2(1 - \alpha)^2]$, and the favored lab wins with probability $(2 - \alpha - c)/[2(1 - \alpha)] > 1/2$. Total compute falls by $(1 - c)V/[2(1 - \alpha)^2]$, and*

$$\Delta J^{\text{excl}} = (1 - \alpha - \mu_N)\,\frac{(1 - c)\,V}{2(1 - \alpha)^2}.$$

(b) Inclusive deal. *Total compute rises by $(1 - c)V/[(c - \alpha)(1 - \alpha)]$, and*

$$\Delta J^{\text{incl}} = -(1 - \alpha - \mu_N)\,\frac{(1 - c)\,V}{(c - \alpha)(1 - \alpha)}.$$

(c) Hence $\operatorname{sign} \Delta J^{\text{excl}} = \operatorname{sign}(1 - \alpha - \mu_N) = -\operatorname{sign} \Delta J^{\text{incl}}$. *If $\alpha + \mu_N < 1$, a supplier and a lab gain by picking a horse and lose by backing the field. If $\alpha + \mu_N > 1$, the reverse holds.*

In both cases the unlinked lab earns zero, so ΔJ is also the coalition's gain.

Proof. Apply Lemma 2 with $c_1 = c$ and $c_2 = 1$ for (a), and Theorem 1 with rebate r for (b). In both cases $\mathbb{E}J = V - (1 - \alpha - \mu_N)T$ by the argument of Theorem 2. Lab b earns zero in the benchmark, as the weak player in the exclusive case, and in the symmetric case. □

The exclusive deal is a pure discouragement device. It does not make the favored lab spend more; it makes the rival spend less and concede more often. Total compute falls, which helps the network precisely when compute is over-supplied, $\alpha + \mu_N < 1$. Inclusive deals do the opposite. The theorem therefore offers a rationale for two observed patterns: some high-margin component suppliers spread their investments across many labs, while cloud providers that also own or sponsor labs tend to concentrate on one. In the model, the choice is governed by a single number, $\alpha + \mu_N$.

Corollary 1 (Vertical integration)**.** *If the supplier fully owns lab a, then $c = 1 - \mu_N$: the integrated lab pays only the real resource cost. The integrated lab wins with probability $(1 + \mu_N - \alpha)/[2(1 - \alpha)]$, and the independent lab's expected spending falls to $(1 - \alpha - \mu_N)V/[2(1 - \alpha)^2]$.*

This is a form of foreclosure. It does not raise the rival's input price; it makes the integrated lab a stronger competitor, which discourages the rival. Unlike in the classic foreclosure model of Ordover et al. (1990), it reduces total input demand. A supplier that sells to everyone may therefore prefer *not* to integrate, unless $\alpha + \mu_N < 1$.

More generally, a supplier may link to both labs on different terms, so that the labs' normalized costs are $c_1 < c_2 \leq 1$. Lemma 2 then gives total expected compute

$$T(c_1, c_2) = \frac{V}{2(c_2 - \alpha)} + \frac{(c_1 - \alpha)V}{2(c_2 - \alpha)^2} = \frac{(c_1 + c_2 - 2\alpha)\,V}{2(c_2 - \alpha)^2}. \tag{13}$$

Proposition 4 (Backing the leader versus the laggard)**.** *For $\alpha < c_1 < c_2$:*

(a) $\partial T/\partial c_1 = V/[2(c_2 - \alpha)^2] > 0$*, so a larger rebate to the* stronger *lab lowers total compute;*

(b) $\partial T/\partial c_2 = -(2c_1 + c_2 - 3\alpha)V/[2(c_2 - \alpha)^3] < 0$*, so a larger rebate to the* weaker *lab raises total compute;*

(c) if the total rebate $r_1 + r_2 = \bar{r}$ is fixed, with $r_1 \geq r_2$, total compute is strictly increasing in the smaller rebate r_2. It is minimized by giving the whole rebate to one lab and maximized by splitting it equally. Hence network value is maximized by the extreme allocation if $\alpha + \mu_N < 1$, and by the equal split if $\alpha + \mu_N > 1$.

Proof. Parts (a) and (b) follow by differentiating (13). For (c), write $c_1 = 1 - \bar{r} + r_2$ and $c_2 = 1 - r_2$. The numerator $c_1 + c_2 - 2\alpha = 2 - \bar{r} - 2\alpha$ does not depend on r_2, while the denominator $2(1 - r_2 - \alpha)^2$ is decreasing in r_2. So T is increasing in r_2 on $[0, \bar{r}/2]$. Combining this with $\mathbb{E}J = V - (1 - \alpha - \mu_N)T$ gives the last claim. □

Proposition 4 explains why the *identity* of the lab a supplier backs matters. Backing the leader entrenches it and calms the race; backing the laggard intensifies the race. When the race is already too intense from the network's point of view, the supplier should back the leader.

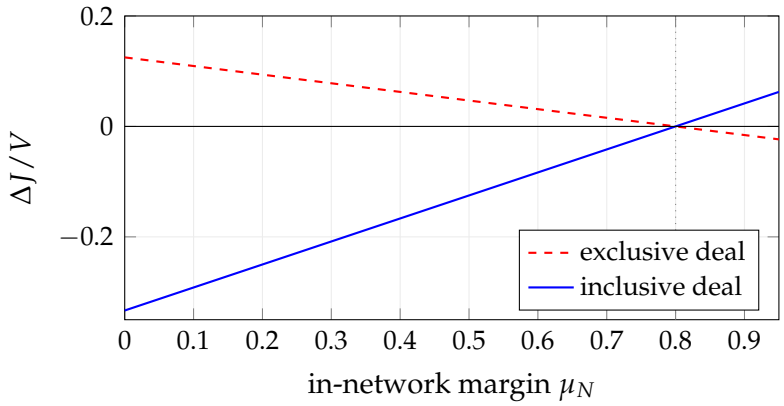


Figure 8: Pick a horse or back the field (Theorem 4), with $\alpha = 0.2$ and $c = 0.8$. The lines cross zero at $\mu_N = 1 - \alpha$ (dotted line).

10 Margins, Fragility, and Busts

Supplier margins in AI compute are high and contested: alternative accelerators, custom silicon and new entrants can compress them. Now we analyze how circularity interacts with margin risk.

Proposition 5 (Fragility). *Fix holdings (σ, q, ϕ_L) with $p = 0$ and $q > 0$, and let the margin μ vary.*

(a) The elasticity of compute spending with respect to the margin is $r/(1 - \alpha - r)$. It is zero without circularity and increasing in r.

(b) The elasticity of the supplier's revenue $2\mu\bar{e}$ with respect to the margin is $(1 - \alpha)/(1 - \alpha - r) > 1$.

(c) *The structure's value* $\Delta J(\mu) = 2(\alpha + \mu - 1)(\bar{e}(\mu) - \bar{e}^0)$ *is positive for* $\mu > 1 - \alpha$ *and negative for* $\mu < 1 - \alpha$. *For any* $\mu < 1 - \alpha$, $|\Delta J(\mu)|$ *is increasing in* q *and* σ.

Proof. By (8), $\ln \bar{e} = \text{const} - \ln(1 - \alpha - q\mu/\theta)$, so $\partial \ln \bar{e} / \partial \ln \mu = r/(1 - \alpha - r)$, which gives (a). Part (b) adds one. For (c), $\bar{e} - \bar{e}^0 > 0$ is increasing in r, and r is increasing in q and σ. □

Proposition 5 formalizes a common worry about circular deals: they make compute demand, and with it supplier revenue, more sensitive to supplier margins. A structure designed when margins are high ($\mu > 1 - \alpha$) creates value at those margins, but if margins fall below $1 - \alpha$, the same structure keeps compute high and destroys value. The more circular the structure, the larger the loss. In Figure 9, with $\sigma = 0.3$ and $q = 0.2$, the structure raises compute by 47% and creates value at $\mu = 0.9$, while at $\mu = 0.6$ it still raises compute by 27% but destroys value.

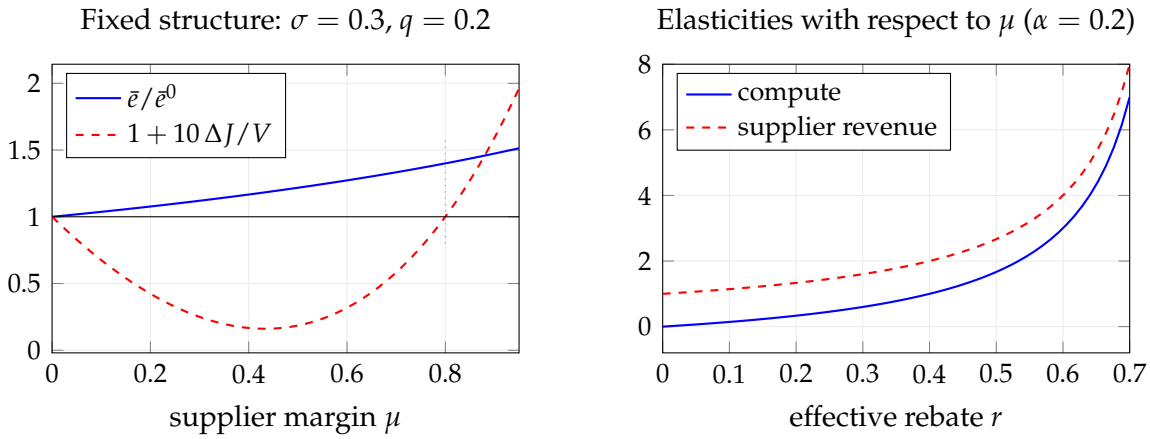


Figure 9: Fragility (Proposition 5). Left: a fixed circular structure creates value only when $\mu > 1 - \alpha = 0.8$ (dotted line). Right: circularity raises the sensitivity of compute and of supplier revenue to the margin.

Remark 1 (Busts). In the model, holdings are fixed when margins change. If unwinding is costly, because equity is illiquid, commitments are binding or warrants have vested, the structure persists after margins fall. The result is a period of over-investment that ends only when the structure is unwound. Spending then falls by the factor $(1 - \alpha - r)/(1 - \alpha)$, and the supplier's revenue falls by more than the margin alone would imply.

11 Designing Circular Deals

So far, holdings were taken as given. Suppose instead that the major investors choose them, together with cash transfers. In practice, the transfers are the cash a supplier pays for its stake or the price at which warrants are granted. Control imposes limits: each lab's major investor must retain at least $\underline{\theta}$ of his lab, and the supplier's major investor must retain at least $\underline{s}$ of the supplier, so $q \leq \bar{q} \equiv (1 - \phi_S - \underline{s})/2$. To keep compute finite, we assume $\bar{q}\mu < \underline{\theta}(1 - \alpha)$.

Proposition 6 (Optimal circularity is bang-bang)**.** *Consider symmetric (inclusive) structures, chosen to maximize the major investors' combined payoff with transfers. Let*

$$\Lambda \equiv \mu(1 - \phi_S) - (1 - \alpha)(1 - \phi_L).$$

(a) The major investors' combined payoff is affine in spending per lab, with slope 2Λ*, for every structure* (σ, q, p)*.*

(b) If $\Lambda > 0$*, the optimal structure maximizes the rebate:* $p = 0$*,* $q = \bar{q}$ *and* $\theta = \underline{\theta}$*, so that* $\sigma = 1 - \underline{\theta} - \phi_L$*. Compute is then* $\bar{e}^* = V/[2(1 - \alpha - \bar{q}\mu/\underline{\theta})]$*.*

(c) If $\Lambda < 0$*, the optimal structure has no vertical stake (*$q = 0$*) and maximizes the horizontal holding* p*: the major investors want* less *compute.*

(d) With Nash bargaining between the supplier's major investor and the labs' major investors, each side's gain equals half of $2\Lambda(\bar{e}^* - \bar{e}^0)$ *in case (b), net of the transfer that implements the split.*

Proof. (a) Firm profits depend on the structure only through $\bar{e}$: $\mathbb{E}\Pi_i = V/2 - (1 - \alpha)\bar{e}$ and $\mathbb{E}\Pi_S = 2\mu\bar{e}$. Hence $J_{\text{ins}} = \text{const} + 2\Lambda\bar{e}$. (b), (c) By (8), $\bar{e}$ is increasing in $r = q\mu/\theta$ and decreasing in p, and its supremum is at the corner stated. (d) Transfers are lump sum, so Nash bargaining splits the surplus equally. □

The proposition has a sharp prediction: circular structures should be either absent or pushed to the limits that control and governance allow. Since Λ falls with margin compression and rises with outside financing of labs, the model predicts waves: circular deals appear when supplier margins are high and lab funding is abundant, and they disappear when margins are compressed.

In the model, the stake the supplier acquires is worth the labs' expected profits, which become negative once circularity raises spending (Proposition 2). A supplier that pays

cash for such a stake is therefore not buying a profitable claim. It is paying for the increase in its own margin revenue that the structure induces. Part of the supplier's revenue is thus funded by its own equity investment, and the rest by the labs' outside investors. This is the precise sense in which the arrangement is "circular".

12 Many Labs

The two-lab race keeps the algebra simple, but the main formulas extend to $n \geq 2$ labs. Let the winner earn $V + \alpha \sum_{k=1}^{n} e_k$, let each lab's major investor hold θ of his lab and q of the supplier, and let the supplier's major investor hold σ of each lab, with $p = 0$.

Proposition 7 (Many labs)**.** *In any symmetric equilibrium in which spending has no atom and zero is in the support, expected spending per lab and total spending are*

$$\bar{e}_n = \frac{V}{n\,(1-\alpha-r)}, \qquad T_n = n\bar{e}_n = \frac{V}{1-\alpha-r}, \qquad r = \frac{q\mu}{\theta}.$$

The network-value identity $\mathbb{E}J = V - (1-\alpha-\mu_N)T$ *and the threshold* $\alpha + \mu_N \gtrless 1$ *continue to hold.*

Proof. Each lab wins with probability $1/n$, so $\mathbb{E}\Pi_i = (V + \alpha n\bar{e}_n)/n - \bar{e}_n$ and $\mathbb{E}\Pi_S = \mu n\bar{e}_n$. A major investor's equilibrium payoff equals the payoff from spending zero, which loses with probability one: $q\mu(n-1)\bar{e}_n$. Equating this with $\theta\mathbb{E}\Pi_i + q\mathbb{E}\Pi_S$ gives $\theta V/n = \bar{e}_n[\theta(1-\alpha) - q\mu]$. The value identity follows as in Theorem 2. □

Notice that total compute does not depend on the number of labs. Entry into the frontier race reallocates spending but does not raise it, because rents are fully dissipated whatever the number of contestants. The circularity multiplier $(1-\alpha)/(1-\alpha-r)$ is also independent of n. Circularity, not entry, is what scales the compute boom.

13 Financing Constraints: When One-Way Stakes Matter

Observation 1(a) relies on labs being able to finance any level of spending. Suppose instead that each lab can spend at most its cash B. To keep the algebra simple, we take $\alpha = 0$, and let the major investor's effective cost be $c = 1 - r$ per dollar.

Proposition 8 (Budget caps)**.** *With budget B per lab:*

(a) *If $B \geq V/c$, the cap never binds. If $V/(2c) \leq B < V/c$, there is an equilibrium in which each lab randomizes uniformly with density c/V on $[0, 2B - V/c]$ and places an atom of mass $2(1 - cB/V)$ at B. Expected spending is $V/(2c)$, as without the cap, and rents are zero.*

(b) *If $B < V/(2c)$, both labs spend B with probability one, and each lab's major investor earns $V/2 - cB > 0$ from the race.*

Hence spending per lab is $\min\{B,\ V/(2(1-r))\}$.

Proof. (a) A lab spending $e < 2B - V/c$ wins with probability ce/V, so its payoff is $V \cdot ce/V - ce = 0$. Spending B wins with probability $1 - \frac{1}{2}a$, where $a = 2(1 - cB/V)$ is the atom, which gives $V(1 - \frac{1}{2}a) - cB = 0$. Spending between $2B - V/c$ and B wins with probability $1 - a$ at a strictly higher cost than $2B - V/c$, so it gives a negative payoff. Expected spending equals $V/(2c)$ because rents are zero and the expected prize is $V/2$ per lab. (b) At the profile (B, B), a lab's payoff is $V/2 - cB > 0$; spending less loses for sure, and spending more is infeasible. □

With financing constraints, the two sides of a circular deal play different roles. The supplier's cash, that is, the stake σ, relaxes the cap B and raises spending one-for-one, but only up to the unconstrained level $V/(2(1-r))$. The vertical stake raises that level itself. Roughly speaking, cash moves the floor and rebates move the ceiling. Once labs are well funded, as frontier labs largely are, only the ceiling matters, and Observation 1(a) applies. The value test of Theorem 2 is unchanged, because it is an accounting identity.

14 Data Centers and Power: A Two-Layer Stack

So far, the supplier simply collects a margin. But data centers and the power to run them must be built, and that buildout is itself competitive and subject to spillovers. In this section, we add a second racing layer, which brings the model closer to the two-industry structure of Xiong (2026).

14.1 Setup

There are two *infrastructure* firms, g and h (data-center or power developers), which race to build capacity $k_g, k_h \geq 0$ for an anchor contract. The two layers are linked as follows.

- The lab race has prize $R_L = V_L + \alpha(e_a + e_b) + \beta_L(k_g + k_h)$: capacity raises the value of frontier models, for instance by lowering inference costs.
- The infrastructure race has prize $R_P = V_P + \alpha_P(k_g + k_h) + \beta_P(e_a + e_b)$: lab compute raises the value of capacity contracts, and α_P captures learning and shared grid upgrades.
- A share μ of lab spending is profit for the infrastructure layer, split equally between the two infrastructure firms, because labs use both.

As in Xiong (2026), α and α_P capture intra-industry spillovers and β_L and β_P capture inter-industry spillovers. Each lab's major investor holds θ_L of his lab, p of the rival lab and q of each infrastructure firm. Each infrastructure firm's major investor holds θ_P of his firm, p_P of the rival infrastructure firm and σ of each lab. Closed ownership requires $\theta_L = 1 - p - 2\sigma$ and $\theta_P = 1 - p_P - 2q$. Firm profits are $\Pi_i = R_L \mathbf{1}\{i \text{ wins}\} - e_i$ for labs and $\Pi_g = R_P \mathbf{1}\{g \text{ wins}\} - k_g + \frac{\mu}{2}(e_a + e_b)$ for infrastructure firms.

14.2 Equilibrium and Value

Now each lab's expected winning prize depends on the infrastructure layer's equilibrium investment, and vice versa. As in Xiong (2026), we solve for the expected prizes as a fixed point.

Theorem 5 (Two-layer stack)**.** *Let $m_L = \mathbb{E}e_i$ and $m_P = \mathbb{E}k_g$. Suppose $\theta_L > p$, $\theta_P > p_P$, $\alpha, \alpha_P > 0$,*

$$\theta_L(1-\alpha) - q(\beta_P + \mu) > 0, \qquad \theta_P(1-\alpha_P) - \sigma\beta_L > 0,$$

and let

$$a_L = \frac{\theta_L - p}{2[\theta_L(1-\alpha) - q(\beta_P + \mu)]}, \qquad a_P = \frac{\theta_P - p_P}{2[\theta_P(1-\alpha_P) - \sigma\beta_L]},$$

with $4a_L a_P \beta_L \beta_P < 1$ and $V_L + 2\beta_L a_P V_P > 0$, $V_P + 2\beta_P a_L V_L > 0$. Then both races have equilibria of the form (5), and

$$m_L = \frac{a_L\,(V_L + 2\beta_L a_P V_P)}{1 - 4a_L a_P \beta_L \beta_P}, \qquad m_P = \frac{a_P\,(V_P + 2\beta_P a_L V_L)}{1 - 4a_L a_P \beta_L \beta_P}. \tag{14}$$

The network's value is

$$\mathbb{E}J = V_L + V_P - 2(1 - \alpha - \beta_P - \mu)\,m_L - 2(1 - \alpha_P - \beta_L)\,m_P. \tag{15}$$

Proof. For a lab's major investor, the difference between the winning and the losing payoff is $(\theta_L - p)R_L$. Expected infrastructure capacity enters only through $\mathbb{E}(k_g + k_h) = 2m_P$, so the difference between the expected winning and losing prizes is $\tilde{V}_L = (\theta_L - p)(V_L + 2\beta_L m_P)$, and $d = (\theta_L - p)\alpha$. The major investor's stake q in each infrastructure firm adds $q(\beta_P + \mu)$ per dollar of his lab's spending, because $\Pi_g + \Pi_h = R_P - k_g - k_h + \mu(e_a + e_b)$. Hence $\kappa = \theta_L - p\alpha - q(\beta_P + \mu)$, and Lemma 1 gives $m_L = a_L(V_L + 2\beta_L m_P)$. Similarly, $m_P = a_P(V_P + 2\beta_P m_L)$, since an infrastructure investor's stake σ in each lab adds $\sigma\beta_L$ per unit of his firm's capacity. Solving this system of equations gives (14). Summing firm profits, and using the fact that exactly one firm wins each race, gives (15). □

Corollary 2 (Value thresholds with cross-layer feedback)**.** *Start from the benchmark,* $p = p_P = \sigma = q = 0$.

(a) Labs invest in infrastructure *($q > 0$, $\sigma = 0$). If* $\beta_P + \mu > 0$, *lab compute rises, and network value rises if and only if*

$$\alpha + \mu + \frac{\beta_L \beta_P}{1 - \alpha_P} > 1.$$

(b) Infrastructure invests in labs *($\sigma > 0$, $q = 0$). If* $\beta_L > 0$, *capacity rises, and network value rises if and only if*

$$\alpha_P + \frac{\beta_L(\mu + \beta_P)}{1 - \alpha} > 1.$$

Proof. For (a): with $\sigma = 0$, $a_P = 1/[2(1 - \alpha_P)]$ does not depend on q, so $\Delta m_P = 2a_P\beta_P\Delta m_L$ exactly. Substituting into (15),

$$\Delta J = -2\Delta m_L\left[(1 - \alpha - \beta_P - \mu) + \tfrac{\beta_P(1-\alpha_P-\beta_L)}{1-\alpha_P}\right] = -2\Delta m_L\left[1 - \alpha - \mu - \tfrac{\beta_L\beta_P}{1-\alpha_P}\right].$$

Part (b) is symmetric, with $\Delta m_L = 2a_L\beta_L\Delta m_P$ and $a_L = 1/[2(1 - \alpha)]$. □

Corollary 2 has three implications. First, the threshold for labs' stakes in infrastructure, such as stakes in data-center developers, is the single-stack threshold $\alpha + \mu$ plus a *feedback term* $\beta_L\beta_P/(1 - \alpha_P)$. The intuition behind this term is a feedback effect: more lab compute raises the value of capacity, more capacity raises the value of the frontier, which again makes the labs want to spend more, and the loop is amplified by the infrastructure layer's own spillover. Second, infrastructure firms' stakes in labs pay off only if capacity spills over strongly onto the frontier (β_L large), and then they are valued *through* the margin and demand linkages $\mu + \beta_P$. Third, without margins ($\mu = 0$) and with all spillovers below

one half, as Xiong (2026) assumes ($0 < |\beta| < \alpha < 0.5$), both thresholds fail: no one-sided cross-shareholding raises the network's value, which is the analogue of the impossibility theorem of Xiong (2026).

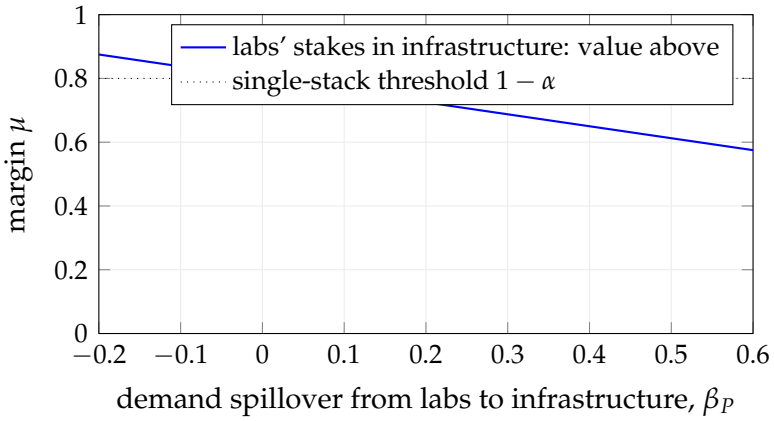


Figure 10: Corollary 2(a) with $\alpha = \alpha_P = 0.2$ and $\beta_L = 0.3$. Cross-layer feedback lowers the margin needed for labs' stakes in infrastructure to create value. With these parameters, infrastructure firms' stakes in labs create value only if $\mu > (1-\alpha_P)(1-\alpha)/\beta_L - \beta_P \geq 1.53$, which is infeasible.

14.3 Power and Leakage

Power is often supplied by independent or regulated producers outside the ownership network. Their margin μ_O is a transfer from the network's perspective, but it is a *leakage*: it is paid by the labs and not recovered by any major investor.

Proposition 9 (Bringing power into the network)**.** *Suppose each lab's major investor acquires a stake q_O in the outside layer, for instance through equity in power projects, in addition to q in the supplier. Then the effective rebate becomes $r = (q\mu + q_O\mu_O)/\theta$, compute is given by (8) with this r, and the value threshold of Theorem 2 becomes $\alpha + \mu + \mu_O > 1$. Integrating power into the network can therefore turn compute-raising circular structures from value-destroying into value-creating.*

Proof. The major investor of lab i gains $q_O\mu_O(e_i + e_j)$, so κ falls by $q_O\mu_O$. Network profits now include Π_O, so the in-network margin in (11) is $\mu + \mu_O$. □

The in-network margin μ_N is thus the key state variable. It is not a property of any single firm: it is the share of a compute dollar that is captured *somewhere* in the ownership network.

15 Discussion and Welfare Analysis

15.1 Private versus Social Thresholds

Let η be the external value of a dollar of compute to parties outside the industry. It is positive if frontier progress diffuses to users and other sectors, and negative if compute imposes grid congestion, emissions or safety risk. A social planner counts all margins as transfers. With total margin $\mu_{\text{tot}} = \mu_N + \mu_{\text{out}}$ (in-network plus outside layers), social welfare changes by

$$\Delta SW = \left(\alpha + \eta + \mu_{\text{tot}} - 1\right) \Delta T. \tag{16}$$

Proposition 10 (Private versus social value of circularity)**.** *Assume a closed network.*

(a) If $\eta \geq 0$, every inclusive circular structure that is privately profitable ($\alpha + \mu_N > 1$) is socially beneficial. There are also socially beneficial structures that are not privately profitable, whenever $\alpha + \mu_N < 1 < \alpha + \eta + \mu_{\text{tot}}$.

(b) If $\eta \geq 0$, exclusive deals and vertical integration that are privately profitable ($\alpha + \mu_N < 1$) are socially harmful whenever $\alpha + \eta + \mu_{\text{tot}} > 1$.

(c) If $\eta < 0$, privately profitable inclusive structures are socially harmful whenever $\alpha + \eta + \mu_{\text{tot}} < 1 < \alpha + \mu_N$.

Proof. Inclusive structures raise T, and exclusive structures lower it (Theorems 1 and 4). The result follows from comparing the signs of $\alpha + \mu_N - 1$ and $\alpha + \eta + \mu_{\text{tot}} - 1$, using $\mu_{\text{tot}} \geq \mu_N$. □

15.2 Policy Implications

Our results have several policy implications.

Antitrust. Exclusive circular deals and vertical integration lower total compute and raise the favored lab's probability of reaching the frontier first (Theorem 4, Corollary 1). They are privately profitable exactly when inclusive deals are not, and they can harm welfare when compute has positive external value. This is a foreclosure channel that works through discouragement in an innovation race rather than through raising rivals' costs. By contrast, inclusive circular structures raise compute and are more likely to be benign.

Disclosure and investor protection. Theorem 3 identifies the labs' outside investors as the party that bears the losses from value-destroying compute booms. Disclosure of labs' stakes in suppliers, of warrants that vest with purchases, and of commitments linked to investments would let outside investors price this transfer.

Energy. Bringing power producers into the ownership network raises compute (Proposition 9). Whether that is desirable depends on η, which for energy-intensive compute includes grid and emissions externalities that the network does not internalize.

Product-market conduct. We have assumed that ownership affects only investment. If cross-holdings also soften competition between the labs once a frontier is reached, there is an additional, conventional anticompetitive effect, as in Reynolds and Snapp (1986). The horizontal holding p captures the investment side of that effect: it lowers compute (Observation 1).

15.3 Modeling Choices

Why a contest? Frontier AI has strong winner-take-most features: being first to a capability level brings users, talent and pricing power, while followers face commoditized markets. An all-pay contest captures this, as well as the fact that compute is sunk whether or not the lab wins. As in Xiong (2026), the contest also delivers full rent dissipation in the benchmark: each lab expects zero profit, which makes every effect of ownership visible in a single number, $\bar{e}$. Xiong (2026) suggests that future work should check whether similar conclusions hold with Cournot competition or Tullock contests. We expect the thresholds to change in such models, because R&D there creates joint surplus that is not fully dissipated. In the present setting, the analogue of that surplus is the margin μ and the cross-layer spillovers, which is why the thresholds here can go either way.

Exogenous margins. If the supplier set its price, and lab spending were measured in dollars, a monopolist would raise its price without bound in this model, because with full dissipation labs spend the same dollars whatever the price. Margins should therefore be read as determined by upstream competition and capacity, which lie outside the model.

Linearity. Prizes and margins are linear in spending. This yields closed forms and makes the value identity (11) exact. With decreasing returns, we expect the thresholds to become

local conditions that hold at the equilibrium rather than globally.

16 An Illustrative Calibration

The model's main thresholds involve two numbers: the knowledge spillover α and the in-network margin μ_N. Table 1 computes μ_N for hypothetical cost shares and margins. The numbers are illustrative assumptions, not estimates; they show how the thresholds would be applied.

Table 1: Illustrative in-network margins

Layer	Share of compute dollar	Margin on that share	Contribution to μ
Accelerators	0.45	0.65	0.29
Data center and cloud	0.40	0.30	0.12
Power	0.15	0.20	0.03

Network contains	μ_N	$\alpha + \mu_N$ ($\alpha = 0.1$)	$\alpha + \mu_N$ ($\alpha = 0.3$)
Accelerator supplier only	0.29	0.39	0.59
Accelerators and cloud	0.41	0.51	0.71
Accelerators, cloud and power	0.44	0.54	0.74

All entries are hypothetical.

Under these assumptions, $\alpha + \mu_N < 1$ in every case. The race over-invests from the network's point of view, and Theorem 2(a) applies: closed-network circular structures that raise compute destroy value, while exclusive deals create it. Two qualifications matter. First, by Theorem 3, compute-raising structures are nevertheless profitable for the major investors when the labs' outside share is large enough. With $\phi_S = 0$, $\alpha = 0.3$ and $\mu_N = 0.41$, the condition is $\phi_L > 1 - 0.41/0.7 \approx 0.41$, a plausible level for labs funded by repeated outside rounds. Second, α is hard to measure: diffusion through distillation, open weights and staff mobility may be large, and Corollary 2 shows that cross-layer feedback adds to it. The regime boundary $\alpha + \mu_N = 1$ is the object that empirical work should target.

17 Testable Implications

The model yields predictions that distinguish its mechanisms from other explanations of circular deals, such as financing constraints or risk sharing.

1. **Vertical stakes, not supplier stakes, move compute** (Observation 1). A supplier's equity in a lab should not, by itself, raise the lab's compute purchases once financing is controlled for. Instruments that pay the lab per unit purchased, such as warrants that vest with deployment, equity in the supplier and volume rebates, should. A financing-constraint explanation predicts the opposite ranking.

2. **Amplification by outside finance** (Observation 1, Proposition 6). Holding the vertical stake fixed, labs whose major investors own a smaller share should respond more strongly to it.

3. **Margin sensitivity** (Proposition 5). The elasticity of a lab's compute spending with respect to its supplier's margin, for example around price cuts or the entry of competing accelerators, should increase with the lab's effective rebate. Suppliers' revenue should react more than one-for-one to margin shocks when their customers are circularly linked.

4. **Accounting losses and outside returns** (Proposition 2, Theorem 3). Labs in more circular structures should report larger operating losses, and their outside investors should earn lower returns, while the total compensation of major investors, including claims on suppliers, need not fall.

5. **Exclusive ties discourage rivals** (Theorem 4). After an exclusive deal, the favored lab's spending should not rise, while rival labs' spending should fall and the favored lab should lead more often.

6. **Who backs the field** (Theorem 4). Suppliers with higher margins on lab spending should be more likely to invest across many labs; lower-margin or vertically integrated investors should concentrate on one.

7. **Power in the network** (Proposition 9). Equity links between AI firms and power producers should be followed by higher compute spending than comparable contract-only arrangements.

18 Conclusion

In this paper, we modelled the race for the AI frontier as an all-pay contest with spillovers, in which the labs buy compute from a supplier that earns a margin, and the labs and the supplier hold shares in each other. Following Xiong (2026), we derived the symmetric

equilibrium investing strategy in closed form and showed that circular holdings raise compute if and only if $q\mu > p(1-\alpha)$. The supplier's stake in a lab is neutral on its own, while the lab's stake in the supplier acts as a volume rebate, and the supplier's stake amplifies it. Unlike in R&D contests across industries, a lab's major investor gains from a compute boom, because the supplier's margin is not competed away. The impossibility theorem reappears at the level of the ownership network: if $\alpha + \mu_N < 1$, no circular arrangement can raise compute and benefit its participants. We also showed that the labs' outside investors pay for value-destroying booms, that exclusive deals and inclusive deals are mirror images, that circularity amplifies margin risk, and that with data centers and power, cross-layer spillovers add feedback terms to the thresholds.

A natural way to extend our model is to make margins endogenous, set by competing suppliers. Another extension is to study dynamic races in which holdings are adjusted over time, which would allow us to model the formation and unwinding of circular structures. Finally, another interesting direction is to estimate α and μ_N, since the regime boundary $\alpha + \mu_N = 1$ determines whether circular investment in the AI stack creates or destroys value.

A Proofs

Proof of Lemma 2. Normalize payoffs so that player k earns $V + \alpha(e_1 + e_2) - c_k e_k$ if it wins and $-c_k e_k$ if it loses. Suppose player 1 follows G_1 while player 2 deviates to investing $s \in [0, \bar{x}]$. Player 2's expected payoff is

$$U_2(s) = \int_{[0,s)} \big(V + \alpha s + \alpha y\big)\, dG_1(y) - c_2 s.$$

Differentiating, $U_2'(s) = g_1(s)\, u(s) + \alpha G_1(s) - c_2$, where g_1 is the p.d.f. of G_1. With $G_1 = (c_2/\alpha)[1 - (V/u)^{1/2}]$, we have $g_1 u = c_2 (V/u)^{1/2}$ and $\alpha G_1 = c_2 - c_2 (V/u)^{1/2}$, so $U_2'(s) = 0$ and U_2 is constant on $[0, \bar{x}]$. Since $G_1(0) = 0$, the constant is $U_2(0) = 0$.

Similarly, suppose player 2 follows G_2, which has an atom at 0, while player 1 deviates to $s \in (0, \bar{x}]$. Then $U_1(s) = \int_{[0,s)} (V + \alpha s + \alpha y)\, dG_2(y) - c_1 s$, and $U_1'(s) = g_2(s) u(s) + \alpha G_2(s) - c_1$, where the atom contributes to $G_2(s)$. With the stated G_2, $g_2 u = \frac{(c_1-\alpha)c_2}{c_2-\alpha}(V/u)^{1/2}$ and $\alpha G_2 = c_1 - \frac{(c_1-\alpha)c_2}{c_2-\alpha}(V/u)^{1/2}$, so $U_1'(s) = 0$. Player 1 never invests exactly 0: investing slightly above 0 beats the atom and earns $V G_2(0)$, which is more than the tie payoff at 0. Hence $U_1 \equiv V G_2(0)$ on $(0, \bar{x}]$.

The upper bound is derived by solving $G_1(\bar{x}) = 1$, which gives $(V/u(\bar{x}))^{1/2} = 1 - \alpha/c_2$, and then $G_2(\bar{x}) = c_1/\alpha - (c_1 - \alpha)/\alpha = 1$. G_2 is increasing, and $G_2(0) = (c_2 - c_1)/(c_2 - \alpha) \in (0, 1)$. Deviating to $s > \bar{x}$ makes a player win with probability one, and its payoff is then strictly decreasing in s since $c_k > \alpha$.

For the means, write $\mathbb{E}e = \int_0^{\bar{x}} [1 - G(x)]\, dx$, with $dx = du/(2\alpha)$. For G_1, the computation is that of Lemma 1 with $\kappa/d = c_2/\alpha$, which gives $V/[2(c_2 - \alpha)]$. For G_2,

$$\mathbb{E}e_2 = \Big(1 - \frac{c_1}{\alpha}\Big)\bar{x} + \frac{(c_1 - \alpha)c_2\sqrt{V}}{\alpha^2(c_2 - \alpha)}\big(\sqrt{u(\bar{x})} - \sqrt{V}\big).$$

Here $\bar{x} = V(2c_2 - \alpha)/[2(c_2 - \alpha)^2]$ and $\sqrt{u(\bar{x})} - \sqrt{V} = \alpha\sqrt{V}/(c_2 - \alpha)$, so $\mathbb{E}e_2 = (c_1 - \alpha)V/[2(c_2 - \alpha)^2]$. Finally,

$$\Pr(e_1 > e_2) = \int G_2\, dG_1 = \frac{c_2\sqrt{V}}{2\alpha}\int_V^{u(\bar{x})} \Big(\frac{c_1}{\alpha} - \frac{(c_1 - \alpha)c_2\sqrt{V}}{\alpha(c_2 - \alpha)}u^{-1/2}\Big)u^{-3/2}\, du,$$

which simplifies to $(2c_2 - c_1 - \alpha)/[2(c_2 - \alpha)]$. This yields the desired result. □

Proof of Corollary 1. With full ownership, the integrated firm's payoff from the lab's spending is $R \cdot \mathbf{1}\{\text{win}\} - e_a + \mu_N(e_a + e_b)$, so its normalized marginal cost is $1 - \mu_N$. Applying Theorem 4(a) with $c = 1 - \mu_N$ gives the result. □

B Numerical Verification

All formulas were verified numerically.

- *Lemma 1.* For 30 random coefficient vectors, the payoff against F is constant on the support (maximum deviation 3×10^{-4}, attributable to the integration grid) and strictly lower above it, and the numerical mean matches (7).
- *Theorems 1 and 2.* Over 20,000 random parameter draws, (9) matches the portfolio payoff and (11) holds, both to machine precision.
- *Lemma 2.* For $\alpha = 0.25$ and $c_1 = 0.7$, $c_2 = 1$: the atom is 0.400; the strong player's payoff varies by less than 10^{-6} on its support; the means are 0.6667 and 0.4000; the win probability is 0.7000; all match the formulas, and both payoffs fall above the support.

- *Corollary 2.* In 19,770 random parameter draws, the sign of ΔJ for one-sided deals agrees with the stated thresholds in every case.

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